“momentum” aren’t properties that an electron has; they are just things we can measure about it. In particular, no particle can have a definite value of both simultaneously. Once we specify the exact wave function for position, the probability of observing any particular momentum is entirely fixed, and vice versa.

The same is true for “vertical spin” and “horizontal spin.”* These are not separate properties an electron can have; they are just different quantities we can measure. If we express the quantum state in terms of the vertical spin, the probability of observing left or right horizontal spin is entirely fixed. The measurement outcomes we can get are determined by the underlying quantum state, which can be expressed in different but equivalent ways. The uncertainty principle expresses the fact that there are different incompatible measurements we can make on any particular quantum state.

Systems with two possible measurement outcomes are so common and useful in quantum mechanics that they are given a cute name: *qubits*. The idea is that a classical “bit” has just two possible values, say, 0 and 1. A qubit (quantum bit) is a system that has two possible measurement outcomes, say, spin-up and spin-down along some specified axis. The state of a generic qubit is a superposition of both possibilities, each weighted by a complex number, the amplitude for each alternative. Quantum computers manipulate qubits in the same way that ordinary computers manipulate classical bits.

We can write the wave function of a qubit as

